Ofqual QAN	**600/0406/X**
Quartz ID	**1156**
Qualification	**Level 4 Diploma in Computing**

Assessment	**Object-Oriented System Analysis and Design**
Assessment Type	**Examination**
Date	**Tuesday 4th March 2025**
Time	**09.00am – 12.00pm**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

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Answer ALL questions

Marks

Question 1

a) Define the following terms in the context of Object-Oriented Analysis and Design:

i) Coupling **1**

Mark scheme
Coupling is the degree to which a module/class is connected to other modules/classes. (1 mark)

ii) Abstraction **2**

Mark scheme
Abstraction is hiding implementation and showing only features. (1 mark)
It allows designers to focus on solving a problem without being concerned about irrelevant lower level details. (1 mark)

iii) Refactoring **2**

Mark scheme
Refactoring is the changing of a software system in such a way that the internal structure is improved (1 mark) without altering the external behaviour or code design. (1 mark)

b) Multiple choice questions: Please select ONE (1) from the options provided. Write your answer in the answer booklet.

i) The Waterfall model can be used in object-oriented development when: **1**

A a working program is required quickly
B the project has a large development team
C requirements are well defined
D legacy systems are developed

Mark scheme
C. Requirements are well defined (1 mark)

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Marks

ii) Unit testing evaluates: **1**

A glass-box testing
B execution of individual methods
C integration of subroutines
D groups of classes that collaborate or communicate in some way

Mark scheme
B. execution of individual methods (1 mark)

iii) Object Constraint Language (OCL) can be used for: **1**

A coding embedded systems
B unit Testing
C ensuring cohesion
D describing pre- and post-conditions on operations and methods

Mark scheme
D. describing pre- and post-conditions on Operations and Methods (1 mark)

iv) In object-oriented design, methods and attributes from a superclass being available to a subclass is called: **1**

A cohesion
B decoupling
C inheritance
D prioritisation

Mark scheme
C. Inheritance (1 mark)

v) In object-oriented design, sufficiency and cohesion can be measured by: **1**

A number of subclasses
B number of responsibilities
C number of attributes
D all of the above

Mark scheme
B. Number of responsibilities (1 mark)

Total 10 Marks

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Marks

Question 2

Identify **and** discuss **FIVE (5) disadvantages** of the Agile Methodology. **10**

Mark scheme
Any five from below or alternative valid answers.

- ***Poor Resource Planning: Allows project to start when requirements are not clear which could result in liability conflict or waste of time on infeasible projects.***
- ***Lack of predictability: Agile is based on the idea that teams will not know the final result (or even a few cycles of delivery down the line) from day one. As a result, it's challenging to predict efforts like cost, time and resources required at the beginning of a project. This challenge becomes more pronounced as projects get bigger and more complex.***
- ***Limited Documentation: In Agile, documentation happens throughout a project, often "just in time" for building the output, rather than at the beginning. Consequently, documentation tends to be less detailed and sometimes falls to the back burner.***
- ***Fragmented Output: Incremental delivery, while helping bring products to market faster, can also lead to fragmented output. When teams work on each component in different cycles, the complete output often lacks cohesion and becomes fragmented.***
- ***No Finite End: Flexibility of requirements can compromise schedule and costs. Agile's minimal planning at the beginning means that projects could have no clear vision of what the "final product" looks like. This lack of a finite end can lead to side-tracking and delivering new, unexpected functionality. Can also result in endless development or huge backlog as new requirements occur or adjustments are requested.***
- ***Difficult Measurement: Agile delivers in increments, making progress tracking challenging. Setting Key Performance Indicators (KPIs) at the start of the project is not always feasible.***
- ***Time Control: Often requires more time for continuous consultation.***

Award 2 marks for each point as 1 mark for identifying each disadvantage and 1 mark for each accompanying explanation.

Total 10 Marks

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Marks

Question 3

a) Explain what is meant by Use-Case Points (UCP) estimation. **4**

Mark scheme
1 mark per valid point + additional mark for elaboration. Eg:

- ***Use-Case Points (UCP) estimation is a method used to estimate the effort required to develop a software system based on its use cases. (1 mark)***
- ***It quantifies the complexity and size of a project, helping in planning and resource allocation. (1 mark)***
- ***Use-Case Points estimation lists all the use cases in the system, which represent the functional requirements. (1 mark)***
- ***UCP considers both the functional requirements and various technical and environmental factors to provide a more accurate effort estimate. (1 mark)***

Alternative response
Developed by Gustav Karner in 1993 and later licensed by Rational Software (1 mark), Use-Case Points provide project size and effort based on number and complexity of use cases, actors. (1 mark)
Complexity is based on non-functional requirements (e.g. performance, portability, maintainability) (1 mark) and Environment (e.g. programming language...) factors. (1mark)

b) Explain **THREE** (3) merits of UCP. **3**

Mark scheme

- ***Based on use cases so measurable early in the project cycle.***
- ***Built on same basis as traditional well-respected models such as Function Point Analysis (FPA) that are able to estimate size independent of skill and experience and then factor these in as part of technical complexity.***
- ***Account for both technical and environmental complexity.***

Award 1 mark for each point up to 3 marks

c) Explain **THREE** (3) challenges of using UCP. **3**

Mark Scheme
Any three from below or equally valid responses.

- ***Dependent on goal-oriented, well-written use cases. If not well or uniformly structured, the resulting UCP may not be accurate.***
- ***Use Case Diagrams need to be drawn before the estimation can commence. There is nothing in the method about the level of granularity of these Use Cases.***
- ***Use cases are goal-oriented and functional in nature. Some work has been done to capture non-functional requirements / technical complexity following Function Point Technical***

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Marks

Complexity tradition but more needs to be considered in terms the varying complexities of the Use Cases.

- ***Use cases do not have a particular size so difficult to use them as a basis for sizing.***
- ***The use cases do not always reflect the actual work that needs to be done in developing interfaces and data.***
- ***Use Case Points do not necessarily capture iterative changes.***

Award 1 mark for each point up to 3 marks

Total 10 Marks

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Marks

Question 4

Compare and contrast object-oriented programming (OOP) with object-based programming (OBP)? **10**

Marking Scheme

- ***Both approaches comprise objects with attributes and values***

Object-Oriented Programming (OOP)

- ***Unlike Object-based languages, object-orientation embodies the concepts of:***
- ***Inheritance: Objects belong to classes that are broken down into subclasses such that an object adopts the attributes and behaviours of its parent class. This promotes code reuse and hierarchical relationships.***
- ***Polymorphism: Different objects can respond to the same message in different ways. This enables flexibility and the ability to define methods that can operate on objects of various types.***
- ***Encapsulation: OOP emphasises encapsulation, where data and methods are bundled together within classes, and access to the data is restricted to ensure data integrity.***
- ***Abstraction: OOP supports abstraction, allowing the creation of complex systems by defining abstract classes and interfaces that specify common behaviours without implementing them.***

Object-Based Programming (OBP)

- ***No Inheritance: OBP does not support inheritance. Objects are created from classes, but there is no mechanism for one class to inherit from another.***
- ***Limited Polymorphism: OBP has limited or no support for polymorphism. Objects are typically treated as instances of their specific classes without the flexibility provided by polymorphism.***
- ***Encapsulation: OBP also supports encapsulation, bundling data and methods within objects. However, the emphasis on encapsulation may not be as strong as in OOP.***
- ***Abstraction: OBP may support some level of abstraction, but it is generally less formalised compared to OOP. Interfaces and abstract classes are not typically used.***
- ***Object-based programming is usually based on an object model that has built-in objects e.g. the Document Object Model (DOM).***

***Award 1 mark per bullet point, up to 10 points*.**

Total 10 Marks

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Marks

Question 5

Compare and contrast State Diagrams **and** Activity Diagrams. **10**

Mark Scheme

- ***State Diagrams and Activity diagrams are both behavioural models. (1 mark)***

- ***State diagrams represent internal or external events that can occur in one or more states. (1 mark) Activity Diagrams began as a variant of State Diagrams, adapted to show connections and dependencies between activities (1 mark).***
- ***The key difference is that State diagrams model state-dependent behaviour and transitions between states (thus also called State Transition Diagrams) (1 mark) whilst Activity Diagrams model the control and flow of activities (UML equivalent of flowcharts) (1 mark). State diagrams focus on modelling state transitions whilst Activity diagrams focus on workflow (1 mark).***
- ***A State Diagram consists of events, states and transitions whilst an Activity Diagram shows activities, actions and transition (1). State Diagrams are therefore typically used to model reactive systems whilst Activity Diagrams are mainly used to model business processes and other non-reactive systems (1).***

- ***Both Diagram types show concurrency and sequential dependency, loops and decisions (1 mark) with Activity Diagrams explicitly using decision points and a State Diagram only doing so sometimes. (1 mark)***

<u>Alternative response</u>: (2 marks for similarity and 8 marks for comparison)

Similarity as explained in first and penultimate points above. (2 marks)

	State Diagram	*Activity Diagram*
Purpose	*Useful for reactive systems (1 mark)*	*Useful for non-reactive systems (1 mark)*
Focus	*Internal state of an object or system (1 mark)*	*Sequence of activities involved in a process or workflow (1 mark)*
Elements	*States, transitions, events (1 mark)*	*Activities, actions, transitions (1 mark)*
Usage examples	*Embedded systems, control systems, real-time systems (1 mark)*	*Business processes, workflows, software processes (1 mark)*

Total 10 Marks

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Marks

Question 6

Demonstrate your understanding of CRC (Class-Responsibility-Collaborator) cards by creating a CRC card for a Patient class for a Hospital System. **10**

Show at least ONE (1) superclass and at least FOUR (4) subclasses that might be associated with a Patient class in a typical Hospital System.

Make any assumptions you believe applicable regarding the responsibilities, collaborators and attributes of the given Patient class.

Mark Scheme

Front of Card

Class Name: Patient	
Superclass: Person ***Subclasses: inPatient, outPatient, femalePatient, malePatient, vitalSignsLog, allergies, contactDetails, EmergencyContact, MedicalHistory***	
Responsibilities: • ***Store and manage patient data*** • ***Retrieve patient record*** • ***Update patient data*** • ***Delete patient record*** • ***Ensure data privacy and security*** • ***Track patient visits and treatments*** • ***…***	***Collaborators:*** ***Appointment*** ***Treatment*** ***HealthcareProvider*** ***…***

Back of Card

Class Description: Class contains patient data as is requirement to manage appointments and treatment. The records are required to ensure continuity of care, for legal and billing purposes. They can also be used for research and quality improvement.
Attributes: ***Patientid: unique reference for each patient*** ***Title: enumerated list of titles*** ***Firstname: string*** ***Lastname: string*** ***Date of Birth: date*** ***Gender: male, female, non-binary…***

Award 1 mark for naming the class and identifying an appropriate superclass
Award 1 mark for up to 1 correct subclass and 2 marks for up to 4 correct subclasses
Award 1 mark for each correct responsibility identified, up to a maximum of 2 marks
Award 1 mark for each correct collaborator identified, up to a maximum of 2 marks

Award 1 mark for the class description

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Marks

Award 1 mark for each correct attribute identified, up to a maximum of 2 marks

Total 10 Marks

Question 7

You have been asked to develop a crop management system for a farmer. The farmer wishes to know which crops are grown in which field, when the crops are planted, harvested and the yield. **10**

The system also needs to keep track of the maintenance of the crops. All records need to include the name of the person performing the activity and its duration.

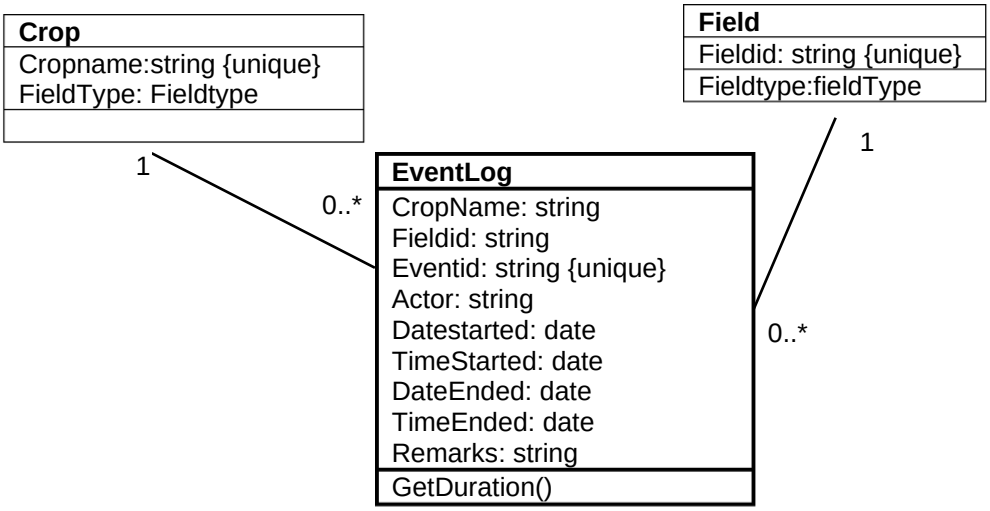
Produce a class diagram showing the following:

- a Crop Class with a Cropname and required FieldType attribute.
- A Field Class with attributes for FieldId and FieldType
- a Crop Management Event class, with
    - crop management event start and end times,
    - person performing the event,
    - comments
    - a method for calculating the duration of the event
- The FieldType is an enumerated field.

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Marks

Mark Scheme



Award 1 mark for correct class and attribute notation

Award 1 mark for Crop attributes

Award 1 mark for Field attributes

Note: Award 1 mark for correctly representing FieldType enumerated type. <u>There is no need to show the types</u> but whilst not penalising add 1 mark if the student has shown this as long as total marks for the question do not exceed 10.

Award 4 marks for EventLog

- ***1 mark for identifier attributes (cropname, Fieldid, eventid, Actor)***
- ***1 mark for start time and end time,***
- ***1 mark for remarks***
  Note: unique identifier can also be a compound index.
- ***1 mark for method.***

Award 3 mark for relationship

- ***Correct relationships with multiplicities (2 marks in total)***
- ***1 mark for correct notation***

Total 10 Marks

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Marks

Question 8

Compare and contrast Communication Diagrams and Sequence Diagrams. **10**

Mark scheme

- ***Communication Diagrams are similar to a Sequence Diagrams in that they both depict messages passed between objects. One could therefore say that they are extensions of Object Diagrams and Use-Case Diagrams (1 mark). They are technically semantically equivalent in that Communication Diagram can be easily converted into Sequence Diagrams and indeed many CASE tools directly generate one from the other (1 mark).***
- ***They are both Interaction Diagrams which means that they are a type of Behavioural Diagrams (1 mark)***
- ***The main difference between them is that Communication Diagrams show a high-level overview of interactions, whilst Sequence Diagrams show a more detailed time ordered views i.e. Communication Diagrams are better than Sequence Diagrams at visualising the relationship between objects collaborating to perform a particular task whilst Sequence Diagrams are better for showing order. (1 mark).***
- ***Communication Diagrams arrange elements according to space whilst in sequence diagrams, the vertical position of messages and activities shows the sequence of events making them a good tool for illustrating time ordering (1 mark)***
- ***Notation differences:***
    - ***Communication Diagrams use a network layout to show connections, whereas Sequence Diagrams use a linear layout to show the sequence of interactions over time (1 mark).***
    - ***Communication diagrams: Objects are depicted by squares, messages passing between objects are depicted by number flow lines. Communication diagrams use message decomposition to show breakdown of messages. (1 mark)***
    - ***Sequence Diagrams: Also show message on flow lines that include decomposition. Objects are also shown as squares and roles as stickmen. (1 mark) Focus of control also called execution occurrence or activation is shown by a thin rectangular lifeline on each object / role that represents the period when the object/role is performing an operation. Each object/role can have many lifelines for different operations. The Diagrams show the lifecycle of objects, including creation and destruction. (1 mark) Constraints are shown by guards. The Sequence Diagram can also be modelled to show duration between operations (1 mark).***

Total 10 Marks

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Marks

Question 9

a) Describe the purpose of using Object Constraint language (OCL) in Unified Modelling Language (UML). Identify **FOUR (4)** advantages it provides in terms of system specification **and** validation. **6**

Mark Scheme
OCL is a formal language that helps specify expressions and restrictions on UML (Unified Modelling Language) models. Using it ensures correctness and consistency of models by adding constraints in a common way (1 mark). It defines signatures and visibility, queries, invariants, initial values, pre and post conditions including calculations. (1 mark)

Advantages of using OCL
(Any four from below or equally valid responses).

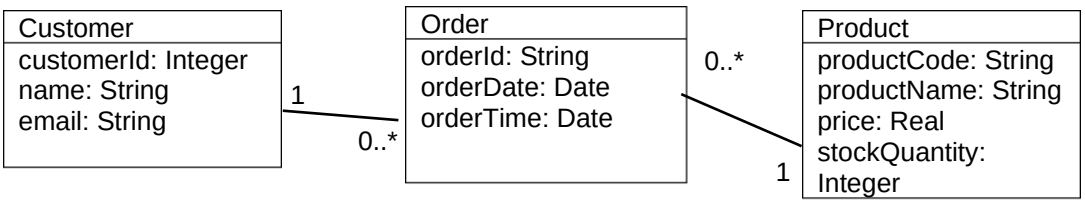
- ***Formal syntax: It provides a way to provide precision and formality to UML diagrams by use of a recognised readable formal syntax.***
- ***Formal reasoning: The formality allows for mathematical reasoning without the need for mathematical skills.***
- ***Declarative: Being declarative means that it describes what should be true rather than how to achieve it. This means it elaborates on the 'how' described in the UML diagrams.***
- ***Maintainable: The formality of the syntax and separation of constraints from models means that it is maintainable.***
- ***Rich query language: It enhances expressiveness and provides a rich query language.***
- ***Side-effect free: It is side-effect free which means that it cannot alter models. It merely explains and clarifies requirements and provides data views.***
- ***Validates and verifies: It can be used to validate and verify UML models to ensure that they adhere to specific rules. It can help identify inconsistencies or error in the early design phase.***
- ***Code generation: It can be used to automatically generate code. It in fact is a key component of Model-Driven Development (MDD) and Model Drien Architecture (MDA).***
- ***Platform independence: It is a high-level language that is independent of any programming language or platform. This means it is applicable across any development project.***

Award 1 mark for each point, up to 4 marks

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Marks

b) Given the UML class diagram for a simple e-commerce application that includes the following classes: **4**



Using OCL, write constraints for the following requirements:

i) A Product must have a price greater than 0.

ii) A Customer must have a valid email address (must contain "@" and "."). The “@” must precede the “.”.

Mark Scheme

(i) Product must have a price greater than 0

context Product inv: self.price > 0 (1 mark)

(ii) Customer must have a valid email address (must contain "@" and "."). The “@” must precede the “.”.

```
context Customer
 inv: self.email.includes('@') and self.email.includes('.')
and self.email.indexOf('@') < self.email.indexOf('.')
```

(Award 1 mark for context; 1 mark for includes; 1 mark for ‘@’ preceding ‘.’

Total 10 Marks

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Marks

Question 10

a) In object-oriented design, explain the difference between Aggregation **and** Composition. **7**

Provide examples to illustrate your answer.

Mark scheme

Aggregation is Whole-Part relationship (part-of/has-a): A class that has component classes objects that are part of a whole (1 mark)

e.g.

- ***arms, legs, head… are part of a body***
- ***or a table has legs or a stand and a table top.***

(1 mark for component, 1 mark for whole)

Can be treated as a unit for some operations. Life of subclasses is independent of whole. (1 mark)

Composition, on the other hand, is a form of aggregation with strong ownership and coincident lifetime (1 mark) i.e. if superclass dies then composite classes also die (1 mark) e.g. if a person dies the arms, legs, head also die (though may not be necessarily valid in this age of transplants where organs remain usable for a period after death). (1 mark)

b) Provide an example of inheritance in the context of object-oriented design. **3**

Mark scheme

Inheritance example: an animal is a living thing as is a plant. (1 mark) All living things breathe, grow, reproduce, respond to stimuli and therefore a plant and an animal both inherit these abilities. (2 marks in total: 1 mark for conceptual understanding, 1 mark for relevance of examples)

Total 10 Marks

END OF PAPER

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Learning Outcomes matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	2	Yes
2	1	Yes
3	2, 4	Yes
4	1, 2, 4	Yes
5	2, 3	Yes
6	2	Yes
7	2, 3	Yes
8	2	Yes
9	1	Yes
10	2	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
1. Discuss various approaches to systems analysis and design and explain their strengths and weaknesses.	Unable to identify and assess approaches.	In an extremely limited way, can identify and assess methodologies and methods to reach limited solutions.	Can adequately identify and assess appropriate methodologies and methods to reach appropriate solutions.	Can sufficiently identify and assess appropriate methodologies and methods to reach good solutions.	Can coherently identify and assess coherent methodologies and methods to reach well-defined and structured solutions.
2. Apply object-oriented techniques for systems analysis and modelling of well-defined but complex problems.	Unable to explain key principles or discuss developments.	In an extremely limited way, can explain principles and discuss developments.	Can adequately explain principles and discuss developments.	Has very good grasp of explain principles and discuss developments.	Can coherently discuss developments and explain and adapt principles.
3. Explain and use UML to represent object-oriented models.	Little to no understanding of UML or object-oriented models.	Limited understanding of UML or object-oriented models.	Adequate ability to apply UML with sufficient representations.	Demonstrates very good ability to apply and adapt UML models with a substantial system understanding.	Outstanding ability to apply UML and adapt models in a manner that demonstrates understanding of effectiveness of methods, actions and results.
4. Understand the integration between the design models and implementation.	Lacks awareness of different perspectives or approaches within the area of study.	Has extremely limited awareness of different perspectives or approaches	Has adequate awareness of different perspectives or approaches within the area of study.	Has very good, informed awareness of different perspectives or approaches within the area of	Excellent awareness of different perspectives or approaches within the area of study.

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		within the area of study.	Understands requirements traceability but with some gaps.	study. Understands requirements traceability.	

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